1 Integrated Squared Error

Suppose we want to estimate a possibly multivariate density function g with support Ω. Call our estimator $\hat{g}_\theta$. The integrated squared error (ISE) is given by

$$\underset{\theta}{\operatorname{argmin}} \|\hat{g}_\theta - g\|_2^2 = \underset{\theta}{\operatorname{argmin}} \int_\Omega \hat{g}_\theta^2 - 2\mathbb{E}_g \hat{g}_\theta.$$

The integrals over the support can then simply be evaluated over a set of predefined grid points. We could also assign weights to each of these points, which is used in standard quadrature integration techniques such as Clenshaw-Curtis.

For example, in the bivariate case, we could say that

$$\text{Loss} = \sum_{j,k} w_j w_k \hat{g}_\theta(s_j, s_k)^2 - \frac{2}{n} \sum_{i=1}^{n} \hat{g}_\theta(x_i, y_i)$$

where (s_j, s_k) are fixed grid points and (x_i, y_i) are actual observed data points. The weights (w_j, w_k) can be set according to quadrature integration methods.

Such a loss function can then be optimized for example in a neural network to train the weight parameters.